

Non-symmetry of biconstrained two-step reachability

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Abstract

The biconstrained reachability function $\psi(x, y)$ is the largest fraction of the first part that some vertex of the third part must reach in two steps, over all finite tripartite graphs satisfying minimum-degree conditions in both directions on each adjacent pair. We answer its symmetry question negatively: $|\psi(13/27, 14/27) - \psi(14/27, 13/27)| \geq 1/108$. The proof establishes the exact rational-boundary formula $\psi(1 - \theta, \theta) = 1 - \theta/d(\theta)$, where $d(\theta)$ is a minimum integer degree of a positive rational regular incidence template. An existing seven-type example and its complement bound both relevant integers by four, making equality impossible. Neither exact value nor their ordering is needed or asserted.

1 The question

For $x, y \in (0, 1]$, consider finite simple graphs whose vertex set is partitioned into nonempty stable sets A, B, C , with no A – C edges, and with

$$\begin{array}{llll} |N_B(a)| \geq x|B| & (a \in A), & |N_A(v)| \geq x|A| & (v \in B), \\ |N_C(v)| \geq y|C| & (v \in B), & |N_B(c)| \geq y|B| & (c \in C). \end{array}$$

Such a graph is (x, y) -*biconstrained*. Define

$$\psi(x, y) = \inf_G \max_{c \in C} \frac{|N_A^2(c)|}{|A|}, \quad N_A^2(c) = \{a \in A : N_B(a) \cap N_B(c) \neq \emptyset\},$$

where the infimum runs over all such finite graphs and partitions. Reached vertices are counted once, irrespective of the number of paths. Complete adjacent pairs show that the admissible class is nonempty. No attainment of this infimum is assumed.

Seymour asked whether $\psi(x, y) = \psi(y, x)$ for all parameters [1, p. 47]. The detailed treatment [2] proves symmetry for the corresponding one-direction function ϕ , but explicitly leaves the biconstrained question unresolved after Theorem 2.3. Reversing a particular graph exchanges the two directions but does not by itself equate the universal infima. We prove the following.

Theorem 1. *For the biconstrained function defined above,*

$$\left| \psi\left(\frac{13}{27}, \frac{14}{27}\right) - \psi\left(\frac{14}{27}, \frac{13}{27}\right) \right| \geq \frac{1}{108}.$$

In particular, ψ is not symmetric.

2 An integer invariant on the rational boundary

Fix a rational $\theta \in (0, 1)$. Let $\mathcal{R}_\theta$ consist of finite binary matrices M with nonempty row set V and nonempty column set I , together with strictly positive rational probability vectors $b = (b_v)_{v \in V}$ and $q = (q_i)_{i \in I}$ satisfying

$$Mq = \theta \mathbf{1}_V, \quad M^\top b = \theta \mathbf{1}_I. \quad (1)$$

Require the columns of M to be pairwise distinct; rows may repeat. Set

$$\Delta(M) = \max_{v \in V} \sum_{i \in I} M_{vi}, \quad d(\theta) = \min_{(M, b, q) \in \mathcal{R}_\theta} \Delta(M).$$

The degree Δ counts distinct column types without weights. If $\theta = p/r$ in lowest terms, the r cyclic intervals of length p on r positions, with both weights uniform $1/r$, give distinct columns and degree p . Thus $\mathcal{R}_\theta$ is nonempty and $1 \leq d(\theta) \leq p$. By well-ordering of positive integers, this minimum is realized by a finite template. This is a separate fact from attainment of the graph infimum defining ψ .

Theorem 2. *For every rational $\theta \in (0, 1)$,*

$$\psi(1 - \theta, \theta) = 1 - \frac{\theta}{d(\theta)}. \quad (2)$$

The value on this boundary is attained by a finite graph.

Proof. First take any finite $(1 - \theta, \theta)$ -biconstrained graph. Put

$$m = |B|, \quad F = \max_{c \in C} \frac{|N_A^2(c)|}{|A|}.$$

If $F = 1$, the required lower bound is immediate. Suppose $F < 1$. Every $c \in C$ misses some $a \in A$. Their B -neighborhoods are disjoint and have sizes at least θm and $(1 - \theta)m$. Hence they partition B , with both size bounds attained. In particular, $|N_B(c)| = \theta m$ for every c . Every row has at least $\theta|C|$ neighbors in C . The B - C edge count is exactly $\theta m|C|$, so every row has exactly $\theta|C|$ neighbors.

Group the vertices of C by their distinct B -neighborhoods $T_1, \dots, T_n$. Let $M_{vi} = \mathbf{1}_{v \in T_i}$, let $b_v = 1/m$, and let q_i be the fraction of C in class i . These positive rational weights satisfy (1). For each i , the vertices missed by a c of class i are exactly

$$A_i = \{a \in A : N_B(a) = B \setminus T_i\}.$$

Indeed, disjointness from T_i and the minimum degree force this equality of neighborhoods. The A_i are distinct, disjoint and nonempty. Writing $p_i = |A_i|/|A|$ and $t = \min_i p_i$ gives $F = 1 - t$. For every $v \in B$, all A_i with $v \in T_i$ are nonneighbors of v , so the B -to- A constraint implies

$$\sum_{i: M_{vi}=1} p_i \leq \theta, \quad t\Delta(M) \leq \theta.$$

Additional vertices of A outside these classes are retained; they cannot weaken this inequality. As $\Delta(M) \geq d(\theta)$, $F \geq 1 - \theta/d(\theta)$ for every admissible graph.

For the upper bound choose a template realizing $d = d(\theta)$, with n columns and supports $T_i \subseteq V$. Every support has b -weight θ . Strict positivity of b and distinctness of the columns make the supports pairwise incomparable. Set $t = \theta/d$ and $\delta = 1 - nt$. The residual is nonnegative, since

$$n\theta = \sum_i b(T_i) = \sum_v b_v \sum_i M_{vi} \leq d.$$

Build a weighted graph with row vertices v of weight b_v in B , vertices c_i of weight q_i in C adjacent to T_i , and vertices a_i of weight t in A adjacent to $V \setminus T_i$. If $\delta > 0$, add a vertex a_* of weight δ adjacent to all of B ; if $\delta = 0$, omit it. Each part has weight one and every retained type has positive weight.

Each a_i sees B -weight $1 - \theta$, while a_* sees one. Each row v sees A -weight $1 - t \sum_i M_{vi} \geq 1 - \theta$. Both directions between B and C have neighbor weight exactly θ by (1). Thus all four constraints hold. The vertex c_i misses a_i , but reaches every a_j for $j \neq i$, since $T_i \setminus T_j \neq \emptyset$. It also reaches a_* when present, because $\theta > 0$. Its reached A -weight is therefore exactly $1 - t$.

Finally, replace each weighted type by a positive integer number of twins proportional to its rational weight, using a common denominator within each part. Join entire type pairs according to the weighted adjacency. All middle types survive, so both the degree fractions and the reachability fractions are preserved exactly. This is an ordinary finite simple graph attaining $F = 1 - t$, which proves (2) and attainment. $\square$

3 The asymmetric pair

We use the regular bipartite template from Figure 1 of [2, p. 6; also Section 12, p. 71]. Its matrix and weights are

$$M = \begin{pmatrix} 1 & 0 & 1 & 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 1 \end{pmatrix}, \quad b = \frac{(5, 5, 3, 3, 3, 4, 4)^T}{27}, \quad q = \frac{(4, 4, 3, 3, 3, 5, 5)^T}{27}.$$

Direct multiplication gives $Mq = (13/27)\mathbf{1}$ and $M^T b = (13/27)\mathbf{1}$. The columns are distinct and the maximum row degree is four. The complement $J - M$, where J is the all-ones matrix, has the same positive weights, regularity $14/27$, distinct columns, and maximum row degree four. Hence, with $d = d(13/27)$ and $e = d(14/27)$,

$$d, e \in \{1, 2, 3, 4\}, \quad \psi(14/27, 13/27) = 1 - \frac{13}{27d}, \quad \psi(13/27, 14/27) = 1 - \frac{14}{27e}.$$

Equality would require $13e = 14d$, impossible in this range. For the quantitative gap, if $d = e$ the difference is $1/(27d) \geq 1/108$. If $e > d$, then $13e - 14d = 13(e - d) - d \geq 10$ and $27de \leq 324$. If $d > e$, then $14d - 13e = 14(d - e) + e \geq 15$ with the same denominator bound. Both latter cases give a larger gap. This proves Theorem 1.

Explicit examples and their limits. The construction above applied to M with degree four gives part sizes $(108, 27, 27)$, A -type sizes $(13, 13, 13, 13, 13, 13, 17)$, and B, C -type sizes given

by the numerators of b, q . Every c reaches 95 vertices of A . The complement gives A -type sizes $(14, 14, 14, 14, 14, 14, 14, 10)$ and reaches 94 vertices of A . Thus $\psi(14/27, 13/27) \leq 95/108$ and $\psi(13/27, 14/27) \leq 47/54$. These are upper bounds, not exact evaluations; their inequality alone would not prove asymmetry. The universal boundary formula is indispensable. The true integers d, e , the exact values, and their ordering remain unevaluated here.

4 Prior work, verification, and disclosure

The finite weighted-graph framework and the seven-type matrix are prior work [2], not new constructions of this note. Theorem 12.2 there concerns symmetry when $\psi(x, y)$ attains its minimal value x ; it does not establish general symmetry. The contribution here is the integer boundary formula, with universal lower and finite upper proofs, and its deduction of non-symmetry at an explicit rational pair. A dated literature audit accompanies the supplement. Its search has finite coverage and does not certify absence of all prior work.

The supplement contains the complete original proof records, exact arithmetic checkers, two independent mathematical audits, source identities and a portable verifier. It checks the matrix, both ordinary graphs, all sixteen possible integer gaps and controls for invalid inferences. The theorem is an elementary argument over all finite graphs; finite tests and file hashes supplement rather than replace its proof. No external optimization theorem, numerical approximation, or presumed extremizer for ψ is required.

AI use and review status. AI tools were used extensively to develop and verify the proofs, write and run checking code, search the literature, draft this note, and conduct adversarial reviews. Independent AI review paths are documented in the accompanying research records. This manuscript is an unrefereed preprint and has not received external human peer review.

References

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